

CHRISTLEY OLUBELA

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EDUCATION

B.Sc. Computer Science, University of Ibadan, Nigeria

SSCE, The International School, Olabisi Onabanjo University, Ago-Iwoye

SKILLS

Languages: Rust, Python, TypeScript, JavaScript, Solidity, SQL, R, MQL5

Core Rust and Systems: Trait-based and generic Application Programming Interface (API) design, published crates ([crates.io](#), [docs.rs](#)), multi-crate Cargo workspace architecture, doctested examples, semantic versioning, mdBook documentation sites, swappable backend design, async/await (tokio), zero-copy decoding, Continuous Integration (CI) via GitHub Actions

Blockchain and DeFi: Flash loans (Aave V3, Balancer, Morpho Blue, MakerDAO DssFlash, Kamino Lend, Solend), atomic and cross-DEX and triangular arbitrage, protocol liquidations, Maximal Extractable Value (MEV), mempool monitoring, oracle backrunning (Pyth, Chainlink), delta-neutral hedging, funding-rate arbitrage and carry, Automated Market Maker (AMM) market making, impermanent-loss modeling, ERC-4626 vaults, StableSwap and Curve math, Uniswap V3/V4 (concentrated liquidity, hooks), Liquid Staking Token (LST) depeg, on-chain execution engines, EIP-2930 access lists, Timeboost

Trade Sizing and Execution Optimization: Profit-maximizing sizing against Constant Product Market Maker (CPMM), StableSwap, and Proactive Market Maker (PMM) liquidity curves, multi-hop heterogeneous route optimization (CPMM, StableSwap, PMM, Balancer), golden-section and bisection sizing, real-time streaming sizing across DEX liquidity pools, explicit guarantee-tier modeling on every sizing result

Lending-protocol coverage: Aave V3 (liquidationCall), Morpho Blue (liquidate), Spark, Fluid (Fluid Liquidity Layer), Compound V3 (absorb), Euler V2 (liquidateVault), Kamino Lend (liquidateObligationAndRedeemReserveCollateral), Save (Solend), MarginFi, HyperLend; health-factor monitoring, liquidation-threshold/Loan-to-Value (LTV)/bonus/close-factor modeling, gas break-even and clip-sizing economics, competitor market-share analysis

Chains, EVM and non-EVM: Ethereum mainnet, Arbitrum One, Base, Gnosis, Berachain, Monad, HyperEVM/Hyperliquid (perps, HyperCore, HyperLend), Solana (Kamino, Save, MarginFi, Drift, Ranger/Voltr), Polkadot (ink!)

Blockchain tooling: Foundry (forge, cast, fuzz tests), alloy-rs, revm/REVM simulation, Reth, op-geth, Inter-Process Communication (IPC)/WebSocket nodes, golden-section and bisection sizing, on-chain profit guards, Gnosis Safe, timelocks

Solana-specific: Yellowstone Geyser gRPC, LiteSVM (in-process Solana Virtual Machine simulation), Jito bundles (multi-region), staked QUIC, Solana Address Lookup Tables (ALT), compute-unit budgeting, Pyth pull oracles

MEV and builder infrastructure: Flashbots Relay, Titan Builder, Beaver Build, eth_sendBundle, eth_sendRawTransactionSync, Jito block-engine bundles, block-builder market-share analysis

Backend and Infra: Axum, Actix, FastAPI, tokio, rayon, REST, WebSockets, PostgreSQL/Supabase, MySQL, Redis, SQLite, Docker/Docker Compose, systemd, bare-metal deployment, Prometheus, Grafana, Vercel, Render

Quant and Math: Generalized Autoregressive Conditional Heteroskedasticity (GARCH), stochastic calculus, Monte Carlo simulation, statistical arbitrage, time-series analysis, Bayesian statistics, game theory, volatility modeling, backtesting, basis trading, derivatives pricing, order-flow and volume-profile microstructure

Frontend: React 18, Vite, Next.js, TypeScript, Progressive Web App (PWA) operator dashboards

Practices: circuit breakers and kill switches, gas budgeting, nonce discipline and multi-wallet execution, secrets management (SecretString, chmod 600, cold/hot wallet separation), adversarial and soak and fuzz testing, low-latency and co-location engineering

EXPERIENCE

Rust Engineer, Algorealm Inc. (Remote) | Jan 2026 - Mar 2026

- Built and maintained automated execution infrastructure for delta-neutral derivative strategies on Hyperliquid perpetuals, including real-time monitoring of funding-rate carry and basis risk.
- Engineered algorithmic execution systems in Rust and Python targeting on-chain microstructural inefficiencies and cross-exchange funding-rate divergence.
- Implemented GARCH and stochastic-volatility forecasting modules to drive automated position-sizing and hedge-ratio logic.
- Built mempool-monitoring and on-chain order-flow tooling to surface MEV opportunities and optimize execution timing.
- Owned the engineering of the full strategy lifecycle: signal generation, backtesting harness, paper-trading environment, and live deployment.

Rust Mentor, H.E.R. DAO Rust School (Remote) | Apr 2026 - Present

- Teach systems programming in Rust to emerging Web3 and DeFi builders: ownership and memory models, concurrency patterns for real-time data feeds, and the type system for safe on-chain state management.
- Ground curriculum in live production systems, including published crates with trait-based public APIs and full documentation sites as teaching examples.

Backend Engineer Intern, Beetle (Hybrid) | Mar 2025 - Jun 2025

- Built algorithmic trading bots from scratch using RSI, EMA, ATR, and VWAP signal logic with automated risk management (dynamic stop-loss, position sizing, drawdown controls).
- Engineered AI-powered Telegram automation bots with MySQL backends for real-time data scraping, storage, and retrieval, using an event-driven pipeline architecture analogous to live market-data ingestion.
- Designed and deployed scalable backend workflows for production-grade reliability under live market conditions.

MEV, LIQUIDATION AND DeFi EXECUTION SYSTEMS (Rust and Solidity)

Corvus - Flash-Loan MEV and Liquidation System for Base (Morpho Blue) and Arbitrum (Aave V3) [repo] | Rust, Solidity 0.8.24/Foundry, op-geth IPC, Prometheus/Grafana

- Two-chain liquidation and arbitrage system covering Morpho Blue's 1.89 billion dollars in active debt on Base and Aave V3's WBTC reserve on Arbitrum, backed by a full protocol-research layer mapping contract registries, oracle feeds, and liquidation parameters on both chains.
- Five strategies including cross-DEX arbitrage, triangular arbitrage, and protocol liquidation, modeled against 584 combined annual liquidation events and 90.01 million dollars in combined annual liquidated volume.
- Modeled combined net annual profit projection of 1,473,642 dollars (about 122,804 dollars per month) at a 20 percent (Base) and 18 percent (Arbitrum) capture-rate assumption, net of gas, DEX slippage, and MEV builder fees.
- FlashExecutor contract enforces an on-chain profit gate, protocol and router whitelists, and a hardcoded cold-wallet sweep; privileged rights held by an immutable Gnosis Safe/hardware wallet. Dual submission via Flashbots Relay and direct redundant submission to Titan Builder and Beaver Build.

Goshawk - Multi-Protocol Liquidation Engine for Ethereum Mainnet [repo] | Rust, Solidity/Foundry, Reth IPC, revm, Prometheus

- Covers six Ethereum mainnet lending protocols representing 15.9 billion dollars in active borrowed debt against 27.2 billion dollars in supplied Total Value Locked (TVL): Aave V3, Morpho Blue, Spark, Fluid, Compound V3, and Euler V2, each with mapped LTV, liquidation-threshold, bonus, and close-factor parameters.
- Per-protocol break-even economics modeled from real gas consumption, including a gas-regime table setting recommended minimum clip sizes for normal versus crash/cascade-spike conditions.
- Flash-loan capital sourced fee-free from Morpho Blue, Balancer V2, and Spark/Maker DSR Flash, or at 0.05 percent from Aave V3; Chainlink oracle feeds monitored against per-feed deviation and heartbeat thresholds ahead of liquidation eligibility checks.
- Signed bundles submitted through Flashbots Relay (88 percent builder market share), with direct redundant submission to Titan Builder and Beaver Build.

Gyrfalcon - Multi-Protocol Solana Liquidation Engine (Kamino Lend primary, plus Save and MarginFi) [repo] | Rust 1.79+, Yellowstone Geyser gRPC, LiteSVM, Jito bundles, staked QUIC

- Primary target Kamino Lend, modeled at 1.34 billion dollars supplied and 1.03 billion dollars borrowed (76.78 percent utilization); one shared, latency-optimized engine extends fault-isolated adapters to Save and MarginFi.
- Modeled against 12 months of Kamino liquidation history: 613 annual events (about 51 per month), 50.55 million dollars total debt liquidated, and a modeled net annual profit of 367,293 dollars (about 30,608 dollars per month) at a 15 percent capture-rate assumption, net of Jito tips and DEX slippage.
- Every candidate liquidation simulated in-process via LiteSVM before an Address Lookup Table (ALT)-aware, compute-unit-budgeted bundle is dual-submitted over staked QUIC and Jito across four regional block-engine endpoints (Amsterdam, New York, Tokyo, Frankfurt); staged observe-to-live rollout gated on backtested adapter accuracy.
- Shipped with a formal profitability-model whitepaper, security policy, and production-readiness checklist.

Kingfisher - Flash-Loan Arbitrage Bot for Curve Stablecoin Pools on Arbitrum One [repo] | Rust (alloy-rs, tokio, rayon), Solidity/Foundry, Aave V3, Curve, systemd

- Borrows stablecoins from Aave V3, captures the StableSwap price spread between two imbalanced Curve pools, repays atomically in one transaction; on-chain profit guard means a losing race reverts and costs only gas.
- Exploits Arbitrum's single-sequencer, no-public-mempool design; broadcasts signed transactions straight to the lowest-latency sequencer, with ordering via Arbitrum Timeboost.
- Co-located near sequencer with a local Nitro node over IPC (about 0.1 millisecond block latency); live-tunable profit floor and sizing cap via a runtime parameters endpoint, without restart.

Kestrel - MEV Arbitrage on Yield-Bearing Stablecoin Mispricings (Ethereum, Arbitrum, Gnosis) [repo] | Rust, Solidity/Foundry, Reth IPC, revm, SQLite, WireGuard

- Monitors the spread between a DEX pool price and the on-chain ERC-4626 redemption rate of yield vaults; executes atomically within one block via flash loan when spread exceeds gas plus fees.
- Five strategies across three chains, including a MakerDAO flash-mint variant and a cross-venue arbitrage strategy; per-strategy profit floors that scale with live gas.
- Local revm simulation before submission, optimal flash-loan sizing via bisection search, multi-builder bundle submission, and a daily gas-budget ceiling with a SQLite profit-and-loss ledger.

Skua - Atomic Flash-Loan Arbitrage System for HyperEVM (Hyperliquid L1, Chain ID 999) [repo] | Rust (async/tokio), Solidity 0.8.24 (Foundry, Cancun), Docker Compose, Prometheus/Grafana

- Four flash-loan strategies against HyperEVM DEXs, HyperLend, and HyperCore, including EVM-to-Core arbitrage, protocol liquidations, triangular arbitrage, and stablecoin depeg capture.
- Async Rust supervisor over a multi-wallet executor; profit enforced on-chain by Solidity flash-loan callbacks that revert unless the trade nets more than gas plus fees.
- Multi-wallet nonce management, golden-section-search optimal sizing, and circuit breakers including a rolling gas-spend kill switch and per-trade capital cap.

Shrike - Flash-Loan Cross-DEX Arbitrage Bot for Berachain (Chain ID 80094) [repo] | Rust hot path, Solidity execution contract, Foundry

- Watches four Berachain DEXs, flash-borrows, and executes cross-DEX arbitrage atomically through a single owner-gated contract enforcing per-leg slippage and an on-chain profit floor.
- Position optimizer combining AMM binary search and a proactive market-maker pricing model; daily gas-budget circuit breaker and max-daily-loss auto-pause.

Peregrine - Flash-Loan MEV System for the Monad L1 [repo] | Rust (2021), Solidity/Foundry, Morpho flash vault, Uniswap V4, Pyth

- Four atomic, flash-loan-funded strategies: cross-venue arbitrage, Liquid Staking Token depeg arbitrage, flash liquidation, and Pyth oracle backrunning.
- Every opportunity simulated in a forked REVM (50 millisecond timeout) before submission; profit booked only from confirmed on-chain receipts, with an automatic phantom-profit halt when REVM predictions diverge from on-chain results.

PUBLISHED RUST LIBRARIES

Driftbrake - Published Rust Crate for Simulation-to-Reality Drift Detection [crates.io] [docs.rs] [repo] | Rust, REVM, tokio, mdBook, GitHub Actions CI, MIT licensed

- Published, installable crate that pre-flight simulates a transaction through REVM and halts a live trading strategy the moment realized profit drifts from the simulated prediction, before the drift compounds into real capital loss.
- Exposes a small set of chain-agnostic traits with zero REVM or alloy dependency; the concrete REVM simulation backend ships as a separate, opt-in crate, so the simulation engine is swappable rather than hardwired.
- Dual-guard halt policy with thresholds derived from a synthetic sweep plotting false-halt rate against missed-catch rate, reproducible through a dedicated benchmark harness.
- Full documentation stack: an mdBook architecture and whitepaper site, a generated docs.rs API reference, a published security policy, and a quickstart example enforced as a doctest.

- Shipped as a five-crate workspace: [driftbrake](#) (facade), [driftbrake-core](#) (chain-agnostic trait boundaries, 114+ downloads), [driftbrake-reconcile](#) (dual-guard halt policy), [driftbrake-receipt-poller](#) (receipt confirmation and profit booking), and [driftbrake-revm-backend](#) (concrete REVM simulation engine).

Optimal-Sizing - Published Rust Workspace for Profit-Maximizing Trade Sizing [\[repo\]](#) | Rust, Cargo multi-crate workspace, CLI, streaming daemon

- Four-crate published workspace that generalizes the trade-sizing logic embedded in Corvus, Kingfisher, and Skua into standalone, reusable infrastructure, published independently of any single bot or chain.
- [sizing-core](#): profit-maximizing trade sizing against Constant Product Market Maker (CPMM), StableSwap, and Proactive Market Maker (PMM) liquidity curves, with an explicit guarantee tier attached to every result.
- [sizing-router](#): multi-hop, heterogeneous trade-route optimization across CPMM, StableSwap, PMM, and Balancer curve types.
- [sizing-watch](#): real-time streaming sizing daemon that recomputes optimal size continuously against live DEX liquidity pools.
- [sizing-cli](#): command-line interface exposing the sizing engine for standalone use outside a live trading system.

QUANT AND TRADING INFRASTRUCTURE

Osprey - Delta-Neutral Funding-Rate Harvesting on Hyperliquid [\[repo\]](#) [\[demo\]](#) | TypeScript (npm-workspaces monorepo), React/Vite, Hyperliquid SDK, FastAPI, 164 passing tests

- Scans every Hyperliquid perpetual every minute, scores funding-rate opportunities, and runs a portfolio of 20 to 100 simultaneous delta-neutral positions, collecting hourly funding with near-zero directional exposure.
- Modules include a live funding heatmap, a harvest engine, a portfolio engine with open-interest caps and concentration limits, and a regime-detection system.

CERTIFICATIONS

IBM Data Science Professional Certificate; IBM AI Engineering Professional Certificate; IBM AI Developer Professional Certificate; Foundations of Business Intelligence, Google; Artificial Intelligence on Microsoft Azure